Q5 Using conservation of energy,

loss in G.P.E. = gain in K.E. + work done against frictional force

[Turn over

$$mgh = \frac{1}{2}mv^2 + f(200)$$

$$(60)g(50) = \frac{1}{2}(60)(20)^2 + f(200)$$

$$f = 87 \text{ N}$$

Answer: A

Q6 For circular motion at the highest point